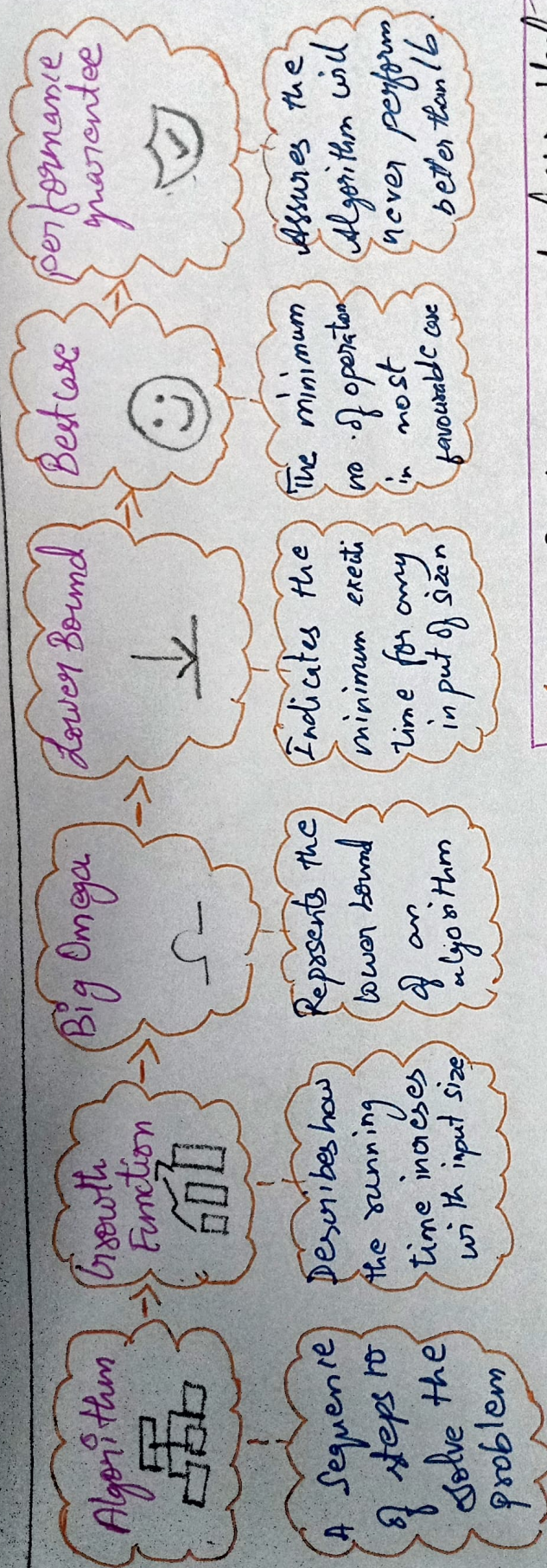
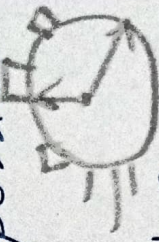


- 1) Concept Map : Algorithm $\rightarrow$ Growth function - Big O $\rightarrow$ Lower Bound $\rightarrow$ Best - Case Explain how Lower bound represent minimum execution time.



Why Best - case Analysis is Useful

* Shows the fastest possible execution time



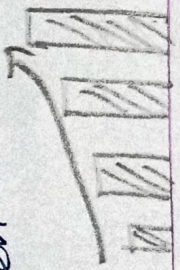
* Helps Evaluate the algorithm under ideal condition

* Provides insight into maximum efficiency of the algorithm

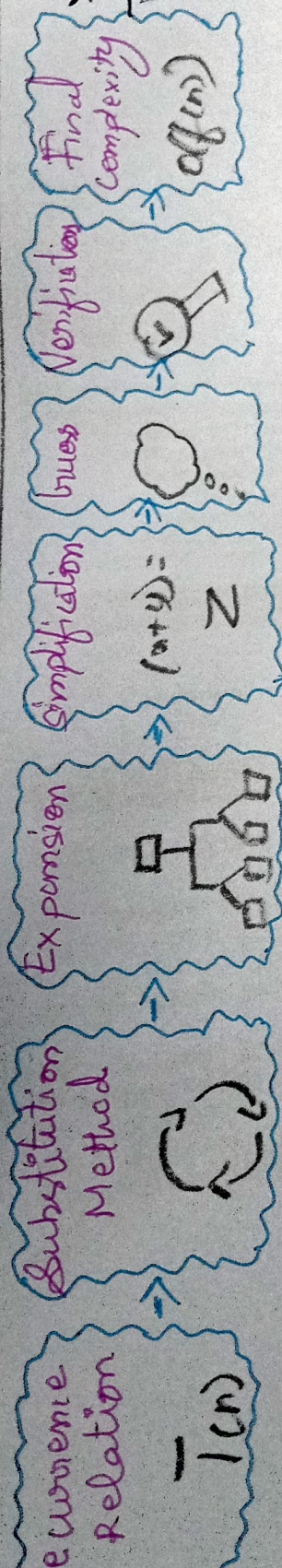
How Lower Bound Represents Minimum Execution Time

Lower Bound (Ω) gives the least number of operations an algorithm must perform.

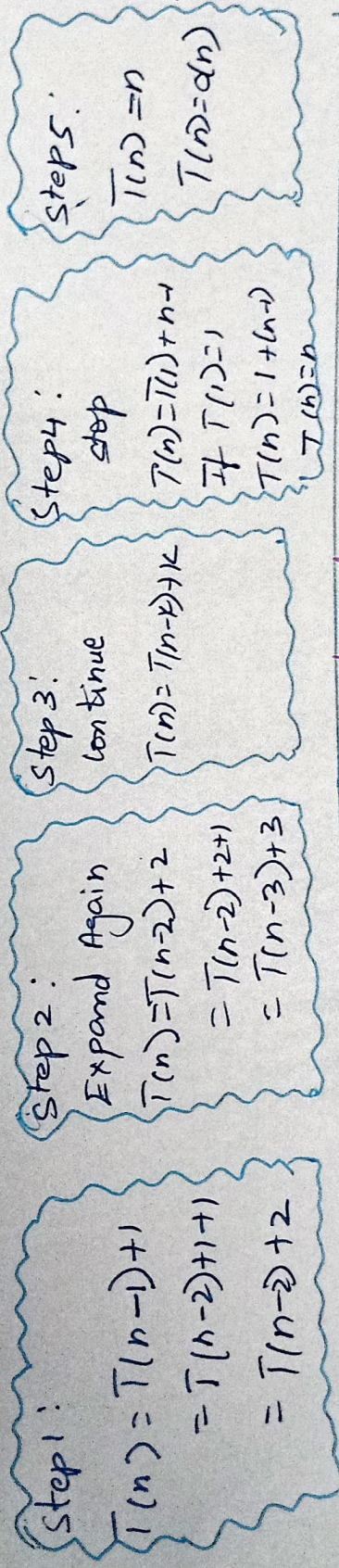
Ex: If $T(n) = \Omega(n)$, then at least n operations are Required even in the Best Case.



2) Concept Map: Recurrence Relation $\rightarrow$ Substitution method
 $\rightarrow$ Expansion $\rightarrow$ Simplification $\rightarrow$ Final Complexity step by
 step transformation of Recurrence into closed form.



step-by-step transformation of Recurrence in closed form $T(n) = T(n-1) + 1$



Interpretation - Benefits of structured approach.

